



TRINETRA

THE THIRD EYE THAT SEES THE DANGER NO SINGLE SENSOR CAN

AI compound-risk intelligence for zero-harm industrial operations

100% compound recall 0% false-positive 7.4 min mean lead

Data present. Unacted upon.

On 13 January 2025, eight workers died
in a coke-oven-battery explosion at the Visakhapatnam Steel Plant.

The gas sensors had data. The permits were logged. SCADA was running.

No layer connected those signals in time.

8

workers killed at Vizag ·
Jan 2025

~3 / day

deaths in India's
registered factories
(DGFASLI)

0

layers connected those
signals in time

The pattern, in a veteran's words

“In more than 30 years in gas and heavy industry, the worst incidents were never one single failure. They were usually three or four small things lining up at the same time, each looking acceptable on its own. Most of the time, nothing actually flagged that combination before it became a serious event.”

NISHAT MULLA · PLANT MANAGER · 30+ YEARS ACROSS GAS & HEAVY INDUSTRY

TRINETRA — OUR RESPONSE

Trinetra is that experience, encoded — it watches every zone 24/7 and flags the combination the moment it forms, so the catch never depends on the right person paying attention at 3 a.m. We enforce safety culture consistently across every shift and handover.

Independent field validation of the problem — a practitioner, in his own words.

Three green lights. One lethal combination.

● RISING FLAMMABLE GAS

still below its alarm

reads NORMAL

● ACTIVE HOT-WORK PERMIT

an ignition source

reads NORMAL

● PERSONNEL PRESENT

inside a confined space

reads NORMAL

A lethal combination no single sensor flags — detected minutes before it becomes critical.

Split reality at T + 8 min

LEGACY SINGLE-SENSOR

● ALL CLEAR

Every gas reads below its setpoint.

threshold detection only

+6

MIN EARLY

TRINETRA COMPOUND AI

● COMPOUND ALERT

COB-1 critical · breach predicted ~36 min
rising CH4 + hot-work permit + 3 personnel

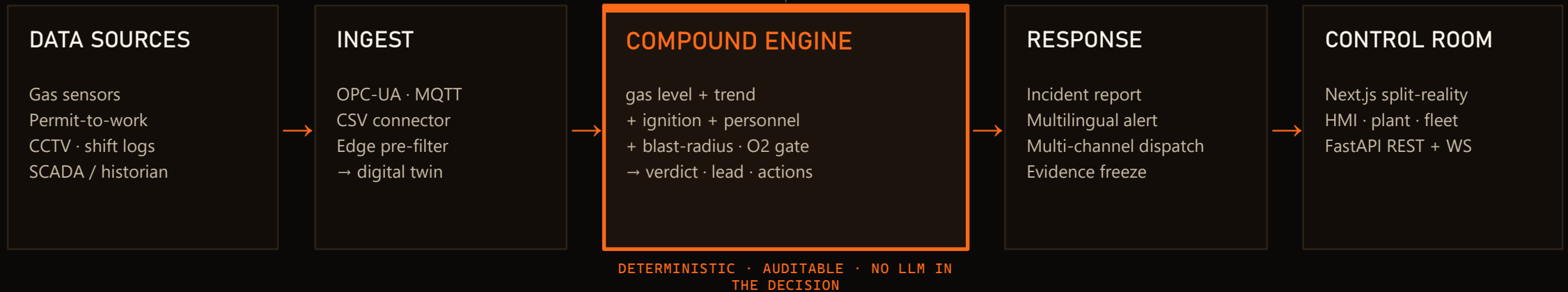
multi-signal fusion

Six minutes of warning — while every gas sensor still reads below its setpoint.

How it works

INTELLIGENCE LAYER · LLM-ASSISTED

Vision (YOLOv8) · Disaster-Memory RAG (Gemini) · 6-stage reasoning graph · Compliance & permit-gate · Pre-mortem · Knowledge graph



Hybrid by design — the deterministic engine makes every life-safety call; the LLM layer only explains it, retrieves precedent, and drafts reports.

We've seen this death before

81

% MATCH

Visakhapatnam coke-oven explosion · 2025 · 8 killed

GROUNDING BRIEFING

"The present condition echoes the Visakhapatnam coke-oven explosion — rising CH₄ alongside an active ignition source mirrors the precedent's root cause of unacted-upon warnings. The single most important action is the immediate cessation of hot-work and evacuation of personnel."

73%

Texas City refinery

71%

Hot-work ignition

—

Piper Alpha

Live conditions embedded with Gemini, matched against a corpus of real industrial disasters.

Find it early. Block it at the desk.

PRE-MORTEM · HAZARDS THAT HAVEN'T HAPPENED YET

228

placements searched

136

compound hazards found

124

cross-zone · walkdowns miss

92

correctly cleared

The engine runs as an oracle across the plant's blast-radius map — surfacing the cross-zone compounds a zone-by-zone walkdown rates safe, before they occur.

SHIFT-LEFT PERMIT GATE · HOT-WORK · COB-1

BLOCKED

HIGH 80



CRITICAL 100 · compound

Simulate the plant WITH the proposed permit; refuse it if issuing it would create — or add people to — a compound hazard. The Vizag hot-work permit, against that atmosphere → BLOCKED.

Two ways to act before the alarm — discover the compound that hasn't formed yet, and refuse the permit that would complete it.

Detect → draft → alert

AUTO-DRAFTED INCIDENT REPORT

- 1 Summary
- 2 Conditions Detected
- 3 Compound Risk Assessment
- 4 Immediate Actions
- 5 Regulatory References
 - Factory Act 1948 §36 / §37 / §38
 - OISD-STD-105 (Work Permit System)
- 6 Corrective Actions

EVACUATION ALERT

Compound hazard in COB-1. Evacuate now. Do not operate hot-work or electrical equipment.

ENGLISH

TELUGU

HINDI

RESPONSE ACTIONS – PREPARED

- ✓ Suspend hot-work permit PTW-7741
- Evacuate & page response team
- Freeze sensor + CCTV evidence
- File preliminary incident report

THE PROOF

Measured, not claimed

100%

compound recall

14 / 14 hazards caught

0%

false-positive

0 / 15 incl. inerted + 02 decoys

7.4

min mean lead

median 6 · max 12 · vs baseline

VIZAG RECONSTRUCTION

+6 MIN LEAD

t0
normal

t8
TRINETRA: COMPOUND

t14
LEGACY: gas alarm

Deterministic (seed=42) · held-out: 100% recall / 3.3% FP over 240 unseen-seed scenarios · property-tested.

— REPLAYED FROM TWO REAL INQUIRIES

Would it catch a real one? Twice.

U.S. CSB · 2005 · Report 2005-04-I-TX

BP TEXAS CITY REFINERY

+10 earlier

min before the documented vapour-cloud ignition (T+20). 15 killed.

MB Lal Committee · 2009

INDIAN OIL JAIPUR DEPOT

+36 earlier

min before the documented ignition (T+48) of a long, undetected vapour build-up. 12 killed.

Same engine, reconstructed inputs. Each inquiry's documented escalation is replayed through the live engine; where a site had no gas detector (a finding in both), the vapour build-up maps to the flammable channel — ignition and personnel are the inquiry's.

The math a buyer underwrites

PER PREVENTED INCIDENT

₹115.5 Cr

avoided loss — a Vizag-class event

Lives protected	₹7.5 Cr
Asset damage	₹40 Cr
Downtime (21 d)	₹63 Cr
Regulatory penalty	₹5 Cr

EXPECTED ANNUAL RETURN

≈ 7.7×

on the ₹1 Cr / plant / yr platform — even at a conservative 1-in-15-year event

3.9×	7.7×	14.4×
1-in-30 yr	1-in-15 yr	1-in-8 yr

+ insurance lever: a recurring ₹0.4–1.2 Cr/yr premium reduction (5–15% for continuous monitoring) offsets the platform cost whether or not an incident ever occurs.

A wedge, a buyer, a moat.

THE PLAY

BUYER

Plant process-safety head — owns permit-to-work + statutory compliance, carries the fatality risk.

WEDGE

Permit-to-work intelligence — one fundable pain — then expand to full gas + permit + CCTV fusion.

PRICING

~₹1 Cr / plant / yr — vs a single downtime-day (~₹3 Cr) or one OISD / NGT penalty.

TRIGGER

Post-Vizag / LG-Polymers pressure — DGMS scrutiny + mandatory near-miss reporting.

WHY AN INCUMBENT CAN'T JUST SHIP IT

Incumbents sell silos

Honeywell / Dräger / MSA sell gas, permits and CCTV SEPARATELY. The danger lives in the seam no one owns.

Deterministic & audit-grade

Compound rules to real OISD / Factory-Act thresholds — domain IP a safety officer can defend in an audit, not a clonable model.

A per-plant data moat

Operator feedback tunes each site's nuisance profile — proprietary data that compounds per site and raises switching cost.

THE ASK

1–2 design partners (steel / refining) for a 90-day pilot on a live permit-to-work + gas feed.

— SCALES HORIZONTALLY · DEPLOYS OVER YOUR STACK

One engine. Every plant. Measured.

100

plants · one engine

\$0.30

per plant / month

~5,000

plants per core

~0.7M

sensor-tags / sec

MEASURED \$/PLANT/MONTH — FALLS WITH SCALE

\$3.00

10 plants

>

\$0.30

100 plants

>

\$0.03

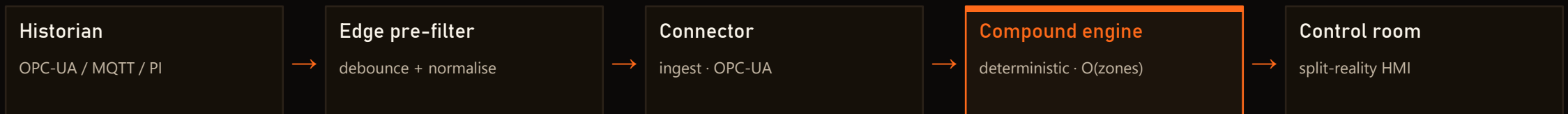
1,000 plants

>

< \$0.01

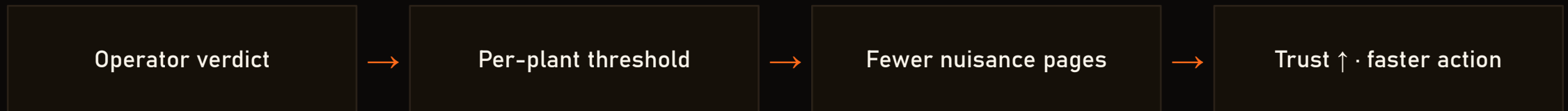
10,000 plants

A CONNECTOR, NOT A REWRITE



Stateless shards — no shared state, no per-site model; the engine is O(zones), so the fleet scales by adding plain workers. Deploys over steel · refining · petrochem · mining · power.

Every plant learns its own nuisance profile



RECALL GUARDRAIL

Compound, HIGH and CRITICAL alerts bypass the threshold entirely. Feedback can only ever damp non-compound nuisance pages — never reduce recall.

THE FLYWHEEL

A confirmed false alarm raises the threshold; a confirmed alert relaxes it. Each plant quietly auto-acknowledges its own routine excursions — fewer nuisance pages, faster action on the ones that matter.

The default engine is untouched — the benchmark (100% recall / 0% false-positive) is byte-identical with or without feedback.



Three workers a day.
The data already exists.

Trinetra is the layer that acts — before, not after.

100% recall 0% false-positive 7.4 min lead